

Communication Complexity

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Overview

In this paper we have studied the information exchange needed, when two processors cooperate to compute Boolean-valued functions.

–Andrew Yao, 1979

Two-party communication complexity (1)

Definition (Two-party communication complexity)

Let $f : \{0, 1\}^{2n} \rightarrow \{0, 1\}$ be a function.

A t -round, two-party protocol Π for computing f is a sequence of t functions $P_1, \dots, P_t : \{0, 1\}^* \rightarrow \{0, 1\}^*$.

An execution of Π on inputs x, y involves the following: at the i th round, if i is odd, then Player 1 computes $p_i = P_i(x, p_1, \dots, p_{i-1})$ and sends p_i to Player 2, and similarly if i is even then Player 2 computes $p_i = P_i(y, p_1, \dots, p_{i-1})$ and sends p_i to Player 1.

Two-party communication complexity (2)

- The protocol Π is valid if for every pair of inputs x, y , the last message sent (i.e., the message p_t) is equal to the value $f(x, y)$.
- The communication complexity of Π is the maximum number of bits communicated (i.e., maximum of $|p_1| + \dots + |p_t|$) over all inputs $x, y \in \{0, 1\}^n$.
- The communication complexity of f , denoted by $C(f)$ is the minimum communication complexity over all valid protocols Π for f .

Goal

Proposition

For every function $C(f) \leq n + 1$

Can they manage with less communication?

Example 1: Parity function

Reminder:

$$\text{PARITY}(x_1, x_2, \dots, x_n, y_1, \dots, y_n) = x_1 \oplus x_2 \oplus \dots \oplus x_n \oplus y_1 \oplus \dots \oplus y_n$$

- Clearly, $C(f) \geq 2$ since the function depends nontrivially on each input
- Consider the following protocol:
Player 1 sends the parity a of the bits in x and Player 2 sends a (XOR) the parity of the bits in y .
Thus $C(f) \leq 2$

So, $C(f) = 2$

Example 2: Halting Problem Variation

Consider the function ¹ $H : \{0, 1\}^n \times \{0, 1\}^n \rightarrow \{0, 1\}$ defined as:

$$H(x, y) = \begin{cases} 1 & \text{if } x = 1^n, y = \text{code}(M) \text{ for some Turing machine } M \text{ s.t. } M \text{ halts on } x \\ 0 & \text{otherwise} \end{cases}$$

Protocol:

- 1 first player sends a bit indicating whether or not his input is 1^n .
- 2 The second player then determines the answer and sends it to the first player.

Thus, the communication complexity of this is at most 2.

¹This example emphasizes that the players have unbounded computational power

Equality Function

We will discuss methods for proving lower bounds on communication complexity.
As a running example, we will use the Equality function:

$$EQ(x, y) = \begin{cases} 1 & \text{if } x = y \\ 0 & \text{otherwise} \end{cases}$$

Theorem (Linear Communication Complexity of Equality)

$$C(EQ) \geq n.$$

We will prove the Theorem using several methods.

The Fooling Set Method: Description

Lemma

Let $f : X \times Y \rightarrow \{0, 1\}$ and let Π be a deterministic protocol for f . If Π produces the same communication pattern (i.e. sequence of bits transmitted) on (x, x) and on (x', x') , then Π produces the same pattern (and hence the same final answer) on

$$(x, x), (x, x'), (x', x), (x', x').$$

Proof. By induction

Lower Bound For Equality

- Assume for contradiction that there is a protocol for EQ with complexity $\leq n - 1$.
- Hence there are at most 2^{n-1} possible communication patterns (transcripts).
- There are 2^n input pairs of the form (x, x) (one for each $x \in \{0, 1\}^n$).
- By the pigeonhole principle, two distinct diagonal pairs (x, x) and (x', x') must share the same transcript.
- By the lemma, that transcript (and therefore the protocol's output) must be the same on the cross-pairs (x, x') and (x', x) as well.
- But $\text{EQ}(x, x) = \text{EQ}(x', x') = 1$ while $\text{EQ}(x, x') = 0$, contradiction.

General Fooling Set Method

Thorem (Generalized Fooling Set Method)

Say that a function $f : \{0, 1\}^n \times \{0, 1\}^n \rightarrow \{0, 1\}$ has a size M fooling set if there is an M -sized subset $S \subseteq \{0, 1\}^n \times \{0, 1\}^n$ and a value $b \in \{0, 1\}$ such that

- (1) for every $\langle x, y \rangle \in S$, $f(x, y) = b$ and
- (2) for every distinct $\langle x, y \rangle, \langle x', y' \rangle \in S$, either $f(x, y') \neq b$ or $f(x', y) \neq b$.

If f has a size- M fooling set then $C(f) \geq \log M$.

Function Matrix & Rectangle Concepts

- **Function Matrix** $M(f)$: Given a function $f : \{0, 1\}^n \times \{0, 1\}^n \rightarrow \{0, 1\}$, its matrix $M(f)$ is a $2^n \times 2^n$ matrix where the $(x, y)^{\text{th}}$ entry is $f(x, y)$.
- **Combinatorial Rectangle**: A *combinatorial rectangle* (or simply *rectangle*) in M is a submatrix corresponding to $A \times B$, where $A, B \subseteq \{0, 1\}^n$.
- **Monochromatic Rectangle**: A rectangle $A \times B$ is called *monochromatic* if $M_{x,y}$ is identical for all $x \in A$ and $y \in B$.

Tiling Method: Idea

- 1 **Start.** Whole matrix $M(f) = \{0, 1\}^n \times \{0, 1\}^n$. Let Π be a Protocol.
- 2 **First bit.** Partition rows into A_0, A_1 where $A_b = \{x | \text{first bit sent} = b\}$.
This yields rectangles $A_0 \times \{0, 1\}^n$ and $A_1 \times \{0, 1\}^n$.
- 3 **Second bit.** Each $A_b \times \{0, 1\}^n$ is refined by $B_{b,0}, B_{b,1}$ where
 $B_{b,d} = \{y : \text{given prefix } b, \text{ player 2 sends } d\}$, producing $A_b \times B_{b,0}$ and $A_b \times B_{b,1}$.
- 4 **Induction.** Each subsequent bit splits every current rectangle into two according to the sender's bit, given the prefix. After t bits the matrix gets partitioned into 2^t rectangles.
- 5 **Transcript view.** Each final rectangle $A_t \times B_t$, equals the set of input pairs producing the same transcript t , hence the same protocol output.

Monochromatic tilings and $\chi(f)$

Definition (monochromatic tiling)

A *monochromatic tiling* of $M(f)$ is a partition of $M(f)$ into disjoint monochromatic rectangles. Denote by $\chi(f)$ the minimum number of rectangles in any monochromatic tiling of $M(f)$.

Theorem (Tiling and communication complexity [AUY83])

$$\log_2 \chi(f) \leq C(f) \leq 16 (\log_2 \chi(f))^2.$$

Proof: The $C(f)$ -bit deterministic protocol has at most $2^{C(f)}$ transcripts, each of which induces a monochromatic rectangle. Hence $\chi(f) \leq 2^{C(f)}$

Fooling Set Lower Bound on Tiling Method

Lemma (Fooling Set Lower Bound)

If f has a fooling set with m pairs, then $\chi(f) \geq m$.

Proof: If (x_1, y_1) and (x_2, y_2) are two of the pairs in the fooling set, then they cannot be in a monochromatic rectangle since not all of $(x_1, y_1), (x_2, y_2), (x_1, y_2), (x_2, y_1)$ have the same f value.

The Rank Method

Now we introduce an algebraic method to lower bound $\chi(f)$

Lemma (Equivalent Rank Definition)

The rank of an $n \times n$ matrix M over a field $\mathbb{F}$, denoted by $\text{rank}(M)$, is the minimum value of ℓ such that M can be expressed as

$$M = \sum_{i=1}^{\ell} B_i$$

where each B_i is an $n \times n$ matrix of rank 1.

Rank Lower Bound

Theorem (Rank Lower Bound)

For every function f , $\chi(f) \geq \text{rank}(M(f))$.

proof: ²

- 1 0 and 1 are elements of every field, although the choice of field can be crucial.
- 2 Every monochromatic rectangle can be viewed (by filling out entries outside the rectangle with 0's) as a matrix of rank at most 1

$$2^{\text{rank}(M(\text{Eq}))} = 2^n \implies C(\text{Eq}) \geq \log \chi(\text{Eq}) \geq n$$

The Discrepancy Method

Definition (Discrepancy)

For this method, we transform f into a ± 1 -valued function via $b \mapsto (-1)^b$ (so $0 \mapsto +1, 1 \mapsto -1$), making $M(f)$ a ± 1 matrix.

The **discrepancy** of a rectangle $A \times B$ in a $2^n \times 2^n$ matrix M is:

$$\frac{1}{2^{2n}} \left| \sum_{x \in A, y \in B} M_{x,y} \right|$$

The **discrepancy of** $M(f)$, denoted $\text{Disc}(f)$, is the maximum discrepancy among all rectangles.

Discrepancy Lower Bound

Discrepancy Lower Bound on Tiling Number

For every function f , $\chi(f) \geq \frac{1}{\text{Disc}(f)}$.

Proof: $\forall k$ s.t. $\chi(f) \leq k$, exists a monochromatic rectangle having at least $2^{2n}/k$ entries.

A rectangle will have discrepancy at least $1/k$.

The inequality holds for the particular $k = \chi(f)$.

Note that this inequality is very loose.

Upper Bounding The Discrepancy

Definition ($\mathcal{E}(f)$)

$$\mathcal{E}(f) = \mathbb{E}_{a_1, a_2, b_1, b_2} \left[\prod_{i=1,2} \prod_{j=1,2} f(a_i, b_j) \right].$$

Lemma (Discrepancy Upper Bound)

$$\text{Disc}(f) \leq \mathcal{E}(f)^{1/4}.$$

Upper Bound Lemma: Proof

We show using simple steps that $\forall$ rectangle $A \times B$, it is true that

$$\mathcal{E}(f) \geq \left(\frac{1}{2^{2n}} \left| \sum_{a \in A, b \in B} f(a, b) \right| \right)^4 = (\mathbb{E}_{a, b \in \{0,1\}^n} [f(a, b)g(a)h(b)])^4$$

where g, h are the characteristic functions of A and B , respectively.

Methods Comparison

Theorem (Discrepancy vs. Rank) [NW94]

For every function f , the discrepancy and rank are polynomially related:

$$\frac{1}{\text{Disc}(f)} = O(\text{rank}(M(f)))$$

Conjecture (Log-Rank)

For every function f , the communication complexity is polynomially related to the logarithm of its rank:

$$C(f) = \text{poly}(\log \text{rank}(M(f)))$$

This would imply that the rank method gives nearly tight bounds on communication complexity.

Multi-party communication complexity

- **Extend players:** from 2 to k players; goal is to compute $f(x_1, \dots, x_k)$.
- **Number-in-hand model:** each player i privately holds x_i (others cannot see x_i); communication is by broadcasts and messages may be adaptive.
- **Number-on-the-forehead (NOF) model — most useful:** player i sees all inputs except x_i (i.e. x_i is “on their forehead”); define $C_k(f)$ as the minimal number of broadcast bits required to compute f .

More Models & Protocols (1/2)

- **Randomized protocols.** Players share a random string; define $R(f)$ as the expected number of bits exchanged. Randomization can drastically reduce cost (e.g., EQ has $O(\log n)$ randomized complexity).
- **Nondeterministic protocols.** Players are given a witness string z ; $f(x, y) = 1$ iff there exists z making the (deterministic) protocol accept. Cost = $|z|$ plus communication; yields NP/coNP analogues in communication complexity.
- **Average-case protocols.** Inputs drawn from distribution D ; define $C_D(f) = \min_P \mathbb{E}_{(x,y) \sim D} [\text{bits exchanged by } P]$, i.e., expected communication under D .

More Models & Protocols (2/2)

- **Computing non-Boolean functions.** Outputs are m -bit values rather than a single bit; protocols/generalizations measure communication to produce multi-bit outputs.
- **Asymmetric communication.** Transmission cost is asymmetric: bits from player 1 cost B times bits from player 2. Objective: minimize the weighted total communication.
- **Computing relations.** Players must agree on any b such that $(x, y, b) \in R$ (a relation); this generalizes function computation to allow multiple valid outputs.

Language as a Communication Problem

Given a language $L \subseteq \{0,1\}^*$, we define the communication complexity problem f_L as follows:

Definition of f_L

For inputs $x, y \in \{0,1\}^*$ with $|x| = |y|$,

$$f_L(x, y) = \begin{cases} 1 & \text{if } xy \in L \\ 0 & \text{otherwise} \end{cases}$$

Streaming Algorithms Connection (1)

Theorem (Communication Complexity and Streaming Algorithms)

If L has a streaming algorithm using $\leq s(n)$ bits of space on all inputs of length $\leq 2n$, then $C(f_L) \leq 4s(n) + 4$.

Streaming Algorithms Connection (2)

Theorem (k -pass Streaming Algorithms Lower Bound)

If L has a k -pass streaming algorithm using $\leq s(n)$ space (in bits), then $C(f_L) \leq O(k \cdot s(n))$.

Streaming Algorithm Lower Bound: Proof

Assume the following protocol:

- Alice runs the deterministic streaming algorithm on x and records the internal memory state m (at most $s(n)$ bits).
- Alice sends a compact encoding of m to Bob (encoding includes a terminator/length marker; overhead $O(s(n))$).
- Bob initializes the streaming algorithm with state m , continues the streaming simulation on y , and outputs whether $xy \in L$.
- Communication cost (compact): encode m with the prefix-free scheme $0 \mapsto 00$, $1 \mapsto 01$ and terminator 11 , until the state is sent, Bob's responds with an arbitrary bit.
- For a k -pass streaming algorithm, repeat the above protocol for each pass — total = $O(k \cdot s(n))$.

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